92

FOR IBPS SBI PO/CLERK PRELIMS 2026

QUANT CHECKLIST

the PRACTICE PAPER

/STUDIFIEDPDF

Subscribe to

STUDIFIED™

Dear Students,

This daily practice sheet is designed to build both speed and accuracy, one day at a time.

It contains a mix of easy, moderate, and challenging questions to prepare you for every possible scenario in the exam. Treat it like a warm-up before the real game.

Solve it daily without fail. Don't wait for motivation—show up with discipline. Because it's not talent but consistent hard work that takes you places.

Stay focused. Stay consistent. Work hard.

-Aashish Arora

PRACTICE PAPER BY AASHISH ARORA


yes OFFICER
ALL COURSES UNDER ONE SUBSCRIPTION'
AASHISH ARORA EXCLUSIVE
MAINS COURSES
MAINS FIRST
THE MATHS HERO PRO
MAINS SETU
MAINS D.I SMARTBOOK
LEVEL UP 1.0, 2.0
MAINS MADE EASY 1.0, 2.0
MAINS SARTHI
OPERATION MAINS
FOUNDATION COURSES
THE MATHS HERO 2026(NEW)
THE MATHS HERO 2025
THE MATHS HERO 2024
UDAAN D.I
GANIT 1.0, 2.0
LAKSHYA COURSE
USE CODE : - TT10
FOR MAXIMUM DISCOUNT
yes OFFICER

CONTENTS

PRACTICE PAPER BY AASHISH ARORA

1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. ? + 37.5% of 720 = ($\sqrt{8281}$ × 14) - 23^2

A. 465

B. 475

C. 485

D. 495

E. None of these

2. 125% of (? - 64) = (13 × 72) - $\sqrt[3]{175616}$

A. 768

B. 756

C. 742

D. 784

E. None of these

3. [$\frac{9}{14}$ of 1078 - $\frac{5}{16}$ of 528] ÷ 11 + 18^2 = ?

A. 382

B. 362

C. 352

D. 372

E. None of these

4. $\sqrt{?}$ + 68.75% of 512 = 26 × 16

A. 3969

B. 4225

C. 4356

D. 4489

E. None of these

5. 15.38% of 1183 + 44.44% of 756 + 7^2 = ? × 7

A. 77

B. 81

C. 79

D. 83

E. None of these

6. 4^3 × 18 - 62.5% of 928 = ? + $\sqrt[3]{110592}$

A. 524

B. 516

C. 520

D. 528

PRACTICE PAPER BY AASHISH ARORA

E. None of these

7. ?% of 800 + 35% of 640 = 56.25% of 960 + 17 × 20

A. 78

B. 84

C. 82

D. 80

E. None of these

8. $2\frac{3}{5}$ of 150 - $1\frac{1}{4}$ of 168 + 75% of 312 = ?

A. 408

B. 418

C. 404

D. 414

E. None of these

9. √(25% of 784 + ?) = 21

A. 255

B. 235

C. 245

D. 225

E. None of these

10. [(36 × 24) - (17 × 32)] ÷ 8 + 31.25% of 576 = ? × 5

A. 42

B. 46

C. 40

D. 44

E. None of these

11. $14^2 + ?^2 = 25^2$ + 125% of 240

A. 29

B. 23

C. 27

D. 25

E. None of these

12. 66.66% of 729 + $\frac{4}{15}$ of 1125 - ? = $13^3 \div 13$

A. 627

B. 617

C. 607

D. 637

E. None of these

13. ($\sqrt{6084} + \sqrt[3]{59319}$) × 6 = ? + 58.33% of 720

A. 278

B. 288

C. 272

D. 282

E. None of these

14. $\frac{3}{17}$ of 2091 + $\frac{7}{18}$ of 972 - $\frac{5}{13}$ of 676 = ?

A. 487

B. 492

C. 477

D. 482

E. None of these

15. ? ÷ 12 + 83.33% of 216 = 44 × 9 - 12^2

A. 852

B. 876

C. 864

D. 888

E. None of these

16. [$\frac{2}{7}$ of 539 + $\frac{5}{8}$ of 528] × 2 - 13^2 = ?

A. 789

B. 809

C. 799

D. 779

E. None of these

17. 225% of 96 + 18.18% of 605 + $\sqrt[3]{91125}$ = ? + 25% of 720

A. 181

B. 191

C. 197

D. 187

E. None of these

18. 13.33% of 975 × 6 - 72^2 ÷ 18 = ?

A. 482

B. 488

C. 492

D. 498

E. None of these

19. $?^2$ - 41.66% of 864 = (27 × 21) + $\sqrt{1156}$

A. 33

B. 29

C. 31

D. 35

E. None of these

20. ($9\frac{1}{3}$ - $4\frac{5}{6}$ + $2\frac{1}{2}$) × 84 = ? + 16^3 ÷ 8

A. 68

B. 76

C. 72

D. 80

E. None of these

Answers:

1. B
2. A
3. D
4. E
5. B
6. A
7. C
8. D
9. C
10. D
11. C
12. B
13. D
14. A
15. C
16. C
17. B
18. C
19. C
20. B

Solutions:

1. ? + 37.5% of 720 = (√8281 × 14) - 23^2

37.5% = $\frac{3}{8}$ and √8281 = 91

So, ? + $\frac{3}{8}$ × 720 = 91 × 14 - 529

? + 270 = 1274 - 529

? + 270 = 745

? = 745 - 270 = 475

2. 125% of (? - 64) = (13 × 72) - $\sqrt[3]{175616}$

125% = $\frac{5}{4}$ and $\sqrt[3]{175616}$ = 56

So, $\frac{5}{4}$ × (? - 64) = 936 - 56

$\frac{5}{4}$ × (? - 64) = 880

? - 64 = 880 × $\frac{4}{5}$

? - 64 = 704

? = 704 + 64 = 768

3. [$\frac{9}{14}$ of 1078 - $\frac{5}{16}$ of 528] ÷ 11 + 18^2 = ?

$\frac{9}{14}$ of 1078 = 693 and $\frac{5}{16}$ of 528 = 165

So, (693 - 165) ÷ 11 + 324 = ?

528 ÷ 11 + 324 = ?

48 + 324 = ?

? = 372

4. √? + 68.75% of 512 = 26 × 16

68.75% = $\frac{11}{16}$

So, √? + $\frac{11}{16}$ × 512 = 416

√? + 352 = 416

CHECKLIST BY AASHISH ARORA

$\sqrt{?} = 416 - 352 = 64$

$? = 64^2 = 4096$

5. 15.38% of 1183 + 44.44% of 756 + 7^2 = ? × 7

$15.38\% = \frac{2}{13}$ and $44.44\% = \frac{4}{9}$

So, $\frac{2}{13} \times 1183 + \frac{4}{9} \times 756 + 49 = ? \times 7$

$182 + 336 + 49 = ? \times 7$

$567 = ? \times 7$

$? = 567 \div 7 = 81$

6. $4^3 \times 18$ - 62.5% of 928 = ? + $\sqrt[3]{110592}$

$4^3 = 64$, $62.5\% = \frac{5}{8}$ and $\sqrt[3]{110592} = 48$

So, $64 \times 18 - \frac{5}{8} \times 928 = ? + 48$

$1152 - 580 = ? + 48$

$572 = ? + 48$

$? = 572 - 48 = 524$

7. ?% of 800 + 35% of 640 = 56.25% of 960 + 17 × 20

$35\% = \frac{7}{20}$ and $56.25\% = \frac{9}{16}$

So, $?\% \times 800 + \frac{7}{20} \times 640 = \frac{9}{16} \times 960 + 340$

$?\% \times 800 + 224 = 540 + 340$

$?\% \times 800 = 880 - 224 = 656$

$? = 656 \times 100 \div 800 = 82$

8. $2\frac{3}{5}$ of 150 - $1\frac{1}{4}$ of 168 + 75% of 312 = ?

$2\frac{3}{5} = \frac{13}{5}$, $1\frac{1}{4} = \frac{5}{4}$ and $75\% = \frac{3}{4}$

So, $\frac{13}{5} \times 150 - \frac{5}{4} \times 168 + \frac{3}{4} \times 312 = ?$

$390 - 210 + 234 = ?$

$414 = ?$

9. $\sqrt{(25\% \text{ of } 784 + ?)} = 21$

$25\% = \frac{1}{4}$

So, $\sqrt{(\frac{1}{4} \times 784 + ?)} = 21$

$\sqrt{(196 + ?)} = 21$

$196 + ? = 21^2 = 441$

$? = 441 - 196 = 245$

10. [(36 × 24) - (17 × 32)] ÷ 8 + 31.25% of 576 = ? × 5

$31.25\% = \frac{5}{16}$

So, $(864 - 544) \div 8 + \frac{5}{16} \times 576 = ? \times 5$

$320 \div 8 + 180 = ? \times 5$

$40 + 180 = ? \times 5$

$220 = ? \times 5$

$? = 44$

11. $14^2 + ?^2 = 25^2$ + 125% of 240

$125\% = \frac{5}{4}$

So, $196 + ?^2 = 625 + \frac{5}{4} \times 240$

$196 + ?^2 = 625 + 300$

$196 + ?^2 = 925$

$?^2 = 925 - 196 = 729$

? = 27

12. 66.66% of 729 + $\frac{4}{15}$ of 1125 - ? = $13^3 \div 13$

$66.66\% = \frac{2}{3}$ and $13^3 \div 13 = 169$

So, $\frac{2}{3} \times 729 + \frac{4}{15} \times 1125 - ? = 169$

486 + 300 - ? = 169

786 - ? = 169

? = 786 - 169 = 617

13. ($\sqrt{6084} + \sqrt[3]{59319}$) × 6 = ? + 58.33% of 720

$\sqrt{6084} = 78$, $\sqrt[3]{59319} = 39$ and $58.33\% = \frac{7}{12}$

So, $(78 + 39) \times 6 = ? + \frac{7}{12} \times 720$

117 × 6 = ? + 420

702 = ? + 420

? = 702 - 420 = 282

14. $\frac{3}{17}$ of 2091 + $\frac{7}{18}$ of 972 - $\frac{5}{13}$ of 676 = ?

$\frac{3}{17}$ of 2091 = 369, $\frac{7}{18}$ of 972 = 378 and $\frac{5}{13}$ of 676 = 260

So, 369 + 378 - 260 = ?

747 - 260 = ?

? = 487

15. ? ÷ 12 + 83.33% of 216 = 44 × 9 - 12^2

$83.33\% = \frac{5}{6}$

So, $? \div 12 + \frac{5}{6} \times 216 = 396 - 144$

? ÷ 12 + 180 = 252

? ÷ 12 = 252 - 180 = 72

? = 72 × 12 = 864

16. [$\frac{2}{7}$ of 539 + $\frac{5}{8}$ of 528] × 2 - 13^2 = ?

$\frac{2}{7}$ of 539 = 154 and $\frac{5}{8}$ of 528 = 330

So, (154 + 330) × 2 - 169 = ?

484 × 2 - 169 = ?

968 - 169 = ?

? = 799

17. 225% of 96 + 18.18% of 605 + $\sqrt[3]{91125}$ = ? + 25% of 720

$225\% = \frac{9}{4}$, $18.18\% = \frac{2}{11}$, $\sqrt[3]{91125} = 45$ and $25\% = \frac{1}{4}$

So, $\frac{9}{4} \times 96 + \frac{2}{11} \times 605 + 45 = ? + \frac{1}{4} \times 720$

$216 + 110 + 45 = ? + 180$

$371 = ? + 180$

$? = 371 - 180 = 191$

18. 13.33% of $975 \times 6 - 72^2 \div 18 = ?$

$13.33\% = \frac{2}{15}$ and $72^2 = 5184$

So, $\frac{2}{15} \times 975 \times 6 - 5184 \div 18 = ?$

$130 \times 6 - 288 = ?$

$780 - 288 = ?$

$? = 492$

19. $?^2 - 41.66\%$ of $864 = (27 \times 21) + \sqrt{1156}$

$41.66\% = \frac{5}{12}$ and $\sqrt{1156} = 34$

So, $?^2 - \frac{5}{12} \times 864 = 567 + 34$

$?^2 - 360 = 601$

$?^2 = 601 + 360 = 961$

$? = 31$

20. $(9\frac{1}{3} - 4\frac{5}{6} + 2\frac{1}{2}) \times 84 = ? + 16^3 \div 8$

$9\frac{1}{3} = \frac{28}{3}$, $4\frac{5}{6} = \frac{29}{6}$ and $2\frac{1}{2} = \frac{5}{2}$

So, $(\frac{28}{3} - \frac{29}{6} + \frac{5}{2}) \times 84 = ? + 4096 \div 8$

$(\frac{56}{6} - \frac{29}{6} + \frac{15}{6}) \times 84 = ? + 512$

$\frac{42}{6} \times 84 = ? + 512$

$7 \times 84 = ? + 512$

$588 = ? + 512$

$? = 76$

CHECKLIST BY AASHISH ARORA

2. ARITHMETIC QUESTIONS

Q1: The length and breadth of a rectangular garden are (x + 7) m and (x − 5) m, respectively. If the cost of levelling the garden at Rs 12 per m^2 is Rs 6480, find the cost of fencing the garden at Rs 18 per meter.
एक आयताकार बगीचे की लंबाई और चौड़ाई क्रमशः (x + 7) मीटर और (x – 5) मीटर है। यदि बगीचे को समतल करने की लागत Rs 12 प्रति m^2 की दर से Rs 6480 है, तो Rs 18 प्रति मीटर की दर से बगीचे की बाड़ लगाने की लागत ज्ञात कीजिए।
(A) Rs 1512
(B) Rs 1620
(C) Rs 1728
(D) Rs 1800
(E) Rs 1944

Q2: Two candidates appeared in a test. One candidate scored 24 marks more than the other. If the marks of the higher-scoring candidate are 56% of the total marks scored by both candidates together, find the marks scored by the higher-scoring candidate.
दो उम्मीदवारों ने एक परीक्षा दी। एक उम्मीदवार ने दूसरे से 24 अंक अधिक प्राप्त किए। यदि अधिक अंक प्राप्त करने वाले उम्मीदवार के अंक दोनों उम्मीदवारों के कुल अंकों के 56% हैं, तो अधिक अंक प्राप्त करने वाले उम्मीदवार के अंक ज्ञात कीजिए।
(A) 104
(B) 112
(C) 118
(D) 124
(E) 128

Q3: A sum of Rs x invested at r% per annum simple interest for 10 years becomes 4 times itself. Another sum of Rs (x + 360), invested at (r + 20)% per annum compound interest compounded annually for 2 years, becomes Rs 10125. Find the value of x.
Rs x की राशि 10 वर्षों के लिए r% वार्षिक साधारण ब्याज पर निवेश करने पर अपनी 4 गुना हो जाती है। Rs (x + 360) की दूसरी राशि को (r + 20)% वार्षिक चक्रवृद्धि ब्याज पर 2 वर्षों के लिए निवेश करने पर Rs 10125 प्राप्त होते हैं। x का मान ज्ञात कीजिए।
(A) 3840

(B) 3960
(C) 4260
(D) 4500
(E) 4140

Q4: Three workers P, Q and R can complete a job in 16 days, 20 days and 40 days, respectively. They complete the whole job together and receive Rs 5500 as total wages. Find the difference between the combined wages of P and R and the wages of Q alone.
तीन श्रमिक P, Q और R किसी कार्य को क्रमशः 16 दिन, 20 दिन और 40 दिन में पूरा कर सकते हैं। वे मिलकर पूरा कार्य करते हैं और कुल मजदूरी के रूप में Rs 5500 प्राप्त करते हैं। P और R की संयुक्त मजदूरी तथा Q की मजदूरी के बीच का अंतर ज्ञात कीजिए।
(A) Rs 1500
(B) Rs 1200
(C) Rs 1800
(D) Rs 2000
(E) Rs 2500

Q5: A merchant purchased two bags for a total of Rs 1750. He sold the first bag at a loss of 16% and the second bag at a profit of 16%. If the selling prices of both bags were equal, find the difference between their cost prices.
एक व्यापारी ने दो बैग कुल Rs 1750 में खरीदे। उसने पहले बैग को 16% हानि पर और दूसरे बैग को 16% लाभ पर बेचा। यदि दोनों बैगों के विक्रय मूल्य बराबर थे, तो उनके क्रय मूल्यों के बीच का अंतर ज्ञात कीजिए।
(A) Rs 240
(B) Rs 320
(C) Rs 260
(D) Rs 280
(E) Rs 360

Q6: A shopkeeper marked a jacket 50% above its cost price and sold it for Rs. 840 after allowing two successive discounts of 30% and y%. If he incurred a loss of 16%, find the value of y.
एक दुकानदार ने एक जैकेट का अंकित मूल्य उसके क्रय मूल्य से 50% अधिक रखा और 30% तथा y% की दो क्रमागत छूट देने के बाद उसे Rs. 840 में बेच दिया। यदि उसे 16% की हानि हुई, तो y का मान ज्ञात कीजिए।
(A) 12%

(B) 15%
(C) 20%
(D) 24%
(E) 25%

Q7: The ratio of the speed of a boat in still water to the speed of the stream is 5:2. The boat travels 210 km downstream from P to Q and returns to P in a total of 20 hours. Find the time taken by the boat to cover 125 km in still water.
शांत जल में नाव की गति और धारा की गति का अनुपात 5:2 है। नाव P से Q तक 210 किमी धारा के अनुकूल जाती है और वापस P आती है। पूरी यात्रा में कुल 20 घंटे लगते हैं। शांत जल में 125 किमी दूरी तय करने में नाव को कितना समय लगेगा?
(A) 4 hours
(B) 6 hours
(C) 8 hours
(D) 5 hours
(E) 7 hours

Q8: The average weight of 29 students in a class is 30 kg. When the teacher's weight is also included, the average weight increases by 2 kg. The average weight of the teacher, his wife, and his daughter is 60 kg. If the daughter's weight is 50% less than her mother's weight, find the weight of the teacher's wife.
एक कक्षा में 29 विद्यार्थियों का औसत वजन 30 किग्रा है। जब शिक्षक का वजन भी शामिल किया जाता है, तो औसत वजन 2 किग्रा बढ़ जाता है। शिक्षक, उनकी पत्नी और उनकी बेटी का औसत वजन 60 किग्रा है। यदि बेटी का वजन उसकी मां के वजन से 50% कम है, तो शिक्षक की पत्नी का वजन ज्ञात कीजिए।
(A) 48 kg
(B) 52 kg
(C) 56 kg
(D) 58 kg
(E) 60 kg

Q9: A person leaves Rs. 38400 to be divided among 3 sons, 2 daughters, and 4 nieces. Each daughter receives twice as much as each niece, and each son receives 4 times as much as each daughter. How much does each son receive?
एक व्यक्ति Rs. 38400 को 3 बेटों, 2 बेटियों और 4 भतीजियों में बांटने के लिए छोड़ता है। प्रत्येक बेटी को प्रत्येक भतीजी से दोगुनी राशि मिलती है, और प्रत्येक बेटे को प्रत्येक बेटी से 4 गुना राशि मिलती है। प्रत्येक बेटे को कितनी राशि मिलेगी?

(A) Rs. 8400
(B) Rs. 9600
(C) Rs. 10800
(D) Rs. 9000
(E) Rs. 10200

Q10: The speed of a boat in still water is 72 km/hr. If the upstream speed of the boat is reduced by 20%, it becomes equal to the speed of the current. Find the downstream speed of the boat.
शांत जल में नाव की गति 72 किमी/घंटा है। यदि नाव की धारा के विपरीत गति को 20% कम कर दिया जाए, तो वह धारा की गति के बराबर हो जाती है। नाव की धारा के अनुकूल गति ज्ञात कीजिए।
(A) 104 km/hr
(B) 96 km/hr
(C) 88 km/hr
(D) 112 km/hr
(E) 120 km/hr

Q11: Average of 15 numbers is 90. The average of the first seven numbers is 85, and the average of the last five numbers is 99. If the ninth number is 25% more than the eighth number and the tenth number is 20% less than the ninth number, then find the ninth number.
15 संख्याओं का औसत 90 है। पहली सात संख्याओं का औसत 85 है और अंतिम पांच संख्याओं का औसत 99 है। यदि नौवीं संख्या, आठवीं संख्या से 25% अधिक है और दसवीं संख्या, नौवीं संख्या से 20% कम है, तो नौवीं संख्या ज्ञात कीजिए।
(A) 92
(B) 100
(C) 96
(D) 108
(E) 112

Q12: Priyansh and Tarun started a business by investing Rs. 6000 and Rs. 4500, respectively. After 5 months, Priyansh withdrew Rs. 2000 from his investment, while Tarun added Rs. 1500 to his investment. Find Tarun’s share from the total annual profit of Rs. 24500.
प्रियांश और तरुण ने क्रमशः Rs. 6000 और Rs. 4500 निवेश करके एक व्यवसाय शुरू किया। 5 महीने बाद, प्रियांश ने अपने निवेश में से Rs. 2000 निकाल लिए, जबकि तरुण ने

अपने निवेश में Rs. 1500 और जोड़ दिए। Rs. 24500 के कुल वार्षिक लाभ में से तरुण का हिस्सा ज्ञात कीजिए।
(A) Rs. 11600
(B) Rs. 12400
(C) Rs. 14000
(D) Rs. 12900
(E) Rs. 13500

Q13: Mixture P of mango juice and water contains 30% mango juice and the rest water. Then 3x ml of mango juice and x ml of water are added to mixture P, so the ratio of mango juice to water in the final mixture becomes 3:4. If the initial quantity of mixture P is M ml, then find the value of M/x.
मिश्रण P में आम का रस और पानी है, जिसमें 30% आम का रस और शेष पानी है। फिर मिश्रण P में 3x ml आम का रस और x ml पानी मिलाया जाता है, जिससे अंतिम मिश्रण में आम के रस और पानी का अनुपात 3:4 हो जाता है। यदि मिश्रण P की प्रारंभिक मात्रा M ml है, तो M/x का मान ज्ञात कीजिए।
(A) 8
(B) 12
(C) 15
(D) 6
(E) 10

Q14: Ishaan’s present age is 20% more than Mohit’s present age but 50% less than Karan’s present age. If the ratio of Mohit’s age 17 years hence to Karan’s age 6 years ago is 2:3, then find the difference between the present ages of Ishaan and Karan.
ईशान की वर्तमान आयु, मोहित की वर्तमान आयु से 20% अधिक है लेकिन करण की वर्तमान आयु से 50% कम है। यदि 17 वर्ष बाद मोहित की आयु और 6 वर्ष पहले करण की आयु का अनुपात 2:3 है, तो ईशान और करण की वर्तमान आयु के बीच का अंतर ज्ञात कीजिए।
(A) 48 years
(B) 36 years
(C) 42 years
(D) 54 years
(E) 45 years

Q15: The downstream speed of a boat is 20% more than its speed in still water. If the boat covers 144 km upstream and 216 km downstream in 12 hours, then find the total time taken by the boat to cover 72 km upstream and 72 km downstream.

एक नाव की धारा के अनुकूल गति, शांत जल में उसकी गति से 20% अधिक है। यदि नाव 144 किमी धारा के प्रतिकूल और 216 किमी धारा के अनुकूल दूरी 12 घंटे में तय करती है, तो 72 किमी धारा के प्रतिकूल और 72 किमी धारा के अनुकूल दूरी तय करने में नाव द्वारा लिया गया कुल समय ज्ञात कीजिए।

(A) 5 hours

(B) 6 hours

(C) 7 hours

(D) 8 hours

(E) 4 hours

Q16: p years from now, the ratio of the ages of Nikhil and Saurabh will be 10:7. After (p + 5) years from now, the ratio of their ages will be 15:11. If the present age of Nikhil is 38 years, then find the present age of Saurabh.

p वर्ष बाद निखिल और सौरभ की आयु का अनुपात 10:7 होगा। आज से (p + 5) वर्ष बाद उनकी आयु का अनुपात 15:11 होगा। यदि निखिल की वर्तमान आयु 38 वर्ष है, तो सौरभ की वर्तमान आयु ज्ञात कीजिए।

(A) 22 years

(B) 30 years

(C) 26 years

(D) 24 years

(E) 28 years

Q17: The LCM of two natural numbers is 24 times their HCF. If the product of the two numbers is 21600, then find the maximum possible difference between the two numbers.

दो प्राकृतिक संख्याओं का लघुत्तम समापवर्त्य उनके महत्तम समापवर्तक का 24 गुना है। यदि दोनों संख्याओं का गुणनफल 21600 है, तो दोनों संख्याओं के बीच अधिकतम संभव अंतर ज्ञात कीजिए।

(A) 630

(B) 720

(C) 570

(D) 690

(E) 600

CHECKLIST BY AASHISH ARORA

Q18: A passenger train running at 72 km/hr crosses a signal pole in 12 seconds. It crosses a goods train running in the opposite direction at 54 km/hr in 18 seconds. Find the length of the goods train.

72 किमी/घंटा की चाल से चल रही एक यात्री ट्रेन 12 सेकंड में एक सिग्नल पोल को पार करती है। यह 54 किमी/घंटा की चाल से विपरीत दिशा में चल रही एक मालगाड़ी को 18 सेकंड में पार करती है। मालगाड़ी की लंबाई ज्ञात कीजिए।

(A) 360 metres
(B) 390 metres
(C) 420 metres
(D) 330 metres
(E) 450 metres

CHECKLIST BY AASHISH ARORA

Q19: On selling an article for Rs. 1700, a seller earns a profit which is twice the loss incurred when the same article is sold for Rs. 800. Find the selling price of the article to earn a profit of 40%.

एक वस्तु को 1700 रुपये में बेचने पर विक्रेता को जितना लाभ होता है, वह उसी वस्तु को 800 रुपये में बेचने पर हुई हानि का दोगुना है। 40% लाभ प्राप्त करने के लिए वस्तु का विक्रय मूल्य ज्ञात कीजिए।

(A) Rs. 1480
(B) Rs. 1500
(C) Rs. 1600
(D) Rs. 1560
(E) Rs. 1540

Q20: Prakash is 50% less efficient than Raghav, and Raghav is 20% more efficient than Tushar. Together, they can complete a piece of work in 42 days. If Raghav and Tushar work together for 36 days and then leave, find the time taken by Prakash alone to complete the remaining work.

प्रकाश, राघव से 50% कम कार्यकुशल है और राघव, तुषार से 20% अधिक कार्यकुशल है। तीनों मिलकर एक कार्य को 42 दिनों में पूरा कर सकते हैं। यदि राघव और तुषार 36 दिनों तक मिलकर काम करते हैं और फिर काम छोड़ देते हैं, तो शेष कार्य को अकेले पूरा करने में प्रकाश को कितना समय लगेगा?

(A) 48 days
(B) 64 days
(C) 56 days
(D) 72 days

(E) 60 days

Answers:

1. C
2. B
3. E
4. A
5. D
6. C
7. D
8. E
9. B
10. A
11. B
12. D
13. E
14. C
15. A
16. C
17. D
18. B
19. E
20. B

CHECKLIST BY AASHISH ARORA

Solutions:

1. Option C

Cost of levelling = Rs 6480 and rate = Rs 12 per m².
Area of garden = 6480 ÷ 12 = 540 m².
So, (x + 7)(x − 5) = 540.
By checking factor pair with difference 12, 30 × 18 = 540.
So, x + 7 = 30 and x − 5 = 18.
Length = 30 m and breadth = 18 m.
Perimeter = 2 × (30 + 18) = 96 m.
Cost of fencing = 96 × 18 = Rs 1728.

2. Option B

Let total marks of both candidates together = 100%.
Higher-scoring candidate = 56% of total.
Lower-scoring candidate = 44% of total.
Difference = 56% − 44% = 12% of total.
Given difference = 24 marks.
So, 12% = 24.
100% = 24 × 100 ÷ 12 = 200.
Marks of higher-scoring candidate = 56% of 200 = 112.

3. Option E

Rs x becomes 4 times itself in 10 years at r% simple interest.
So, interest = 4x − x = 3x.
Simple interest = x × r × 10 ÷ 100.
So, x × r × 10 ÷ 100 = 3x.
r/10 = 3.
r = 30%.
Now, rate for compound interest = r + 20 = 50%.
Amount after 2 years = Rs 10125.
So, x + 360 = 10125 × 100/150 × 100/150.
x + 360 = 10125 × 4/9 = 4500.
x = 4500 − 360 = 4140.

4. Option A

Wages are divided in the ratio of efficiency.
Efficiency ratio of P, Q and R = 1/16 : 1/20 : 1/40.
LCM of 16, 20 and 40 = 80.
So, efficiency ratio = 5 : 4 : 2.
Total parts = 5 + 4 + 2 = 11.
Value of 1 part = 5500 ÷ 11 = 500.
P and R together = (5 + 2) parts = 7 parts.
Wages of P and R together = 7 × 500 = Rs 3500.
Wages of Q = 4 × 500 = Rs 2000.
Required difference = 3500 − 2000 = Rs 1500.

5. Option D

Let the cost prices of the first and second bags be A and B.
First bag is sold at 16% loss, so its selling price = 84% of A.
Second bag is sold at 16% profit, so its selling price = 116% of B.
Both selling prices are equal.
So, 84% of A = 116% of B.
A : B = 116 : 84 = 29 : 21.
Total cost price = Rs 1750.
Total parts = 29 + 21 = 50.
Value of 1 part = 1750 ÷ 50 = 35.
Difference between cost prices = (29 − 21) parts = 8 parts.
Required difference = 8 × 35 = Rs 280.

CHECKLIST BY AASHISH ARORA

6. Option C

Selling price = Rs. 840
Loss = 16%
So, selling price = 84% of cost price
Cost price = $840 \times 100/84$ = Rs. 1000
Marked price = 150% of cost price
Marked price = $1000 \times 150/100$ = Rs. 1500
After 30% discount, price = $1500 \times 70/100$ = Rs. 1050
Now, after y% discount, selling price becomes Rs. 840
So, $1050 \times (100 - y)/100 = 840$
$100 - y = 840 \times 100/1050 = 80$
$y = 100 - 80 = 20$

7. Option D

Let the speed of the boat in still water = 5x km/hr
Speed of stream = 2x km/hr
Downstream speed = $5x + 2x = 7x$ km/hr
Upstream speed = $5x - 2x = 3x$ km/hr
Total time = downstream time + upstream time
$20 = 210/7x + 210/3x$
$20 = 30/x + 70/x$
$20 = 100/x$
$x = 5$
Speed of boat in still water = $5x = 5 \times 5 = 25$ km/hr
Time taken to cover 125 km in still water = 125/25 = 5 hours

8. Option E

Total weight of 29 students = $29 \times 30 = 870$ kg
When teacher is included, total persons = 30
New average = 30 + 2 = 32 kg
Total weight of 30 persons = $30 \times 32 = 960$ kg
Teacher's weight = $960 - 870 = 90$ kg
Average weight of teacher, wife, and daughter = 60 kg
Total weight of teacher, wife, and daughter = $3 \times 60 = 180$ kg
Wife + daughter = $180 - 90 = 90$ kg
Let wife's weight = 100 units

Daughter's weight is 50% less than wife's weight, so daughter's weight = 50 units
Total = 150 units = 90 kg
Wife's weight = $90 \times 100/150 = 60$ kg

9. Option B

Let each niece receive Rs. x
Each daughter receives twice of each niece, so each daughter = Rs. 2x
Each son receives 4 times of each daughter, so each son = $4 \times 2x$ = Rs. 8x
Total amount = amount of 4 nieces + amount of 2 daughters + amount of 3 sons
$38400 = 4x + 2 \times 2x + 3 \times 8x$
$38400 = 4x + 4x + 24x$
$38400 = 32x$
$x = 38400/32 = 1200$
Each son receives = $8x = 8 \times 1200$ = Rs. 9600

10. Option A

Let the speed of current be x km/hr
Speed of boat in still water = 72 km/hr
Upstream speed = $72 - x$
When upstream speed is reduced by 20%, it becomes equal to the speed of current.
So, 80% of upstream speed = current speed
$(72 - x) \times 80/100 = x$
$(72 - x) \times 4/5 = x$
$288 - 4x = 5x$
$9x = 288$
$x = 32$ km/hr
Downstream speed = still water speed + current speed
Downstream speed = $72 + 32 = 104$ km/hr

11. Option B

Total of 15 numbers = $15 \times 90 = 1350$.
Sum of first seven numbers = $7 \times 85 = 595$.
Sum of last five numbers = $5 \times 99 = 495$.
So, sum of 8th, 9th and 10th numbers = $1350 - 595 - 495 = 260$.
Let the 8th number = 4x.
Then 9th number = 25% more than 8th = 5x.

10th number = 20% less than 9th = 80% of 5x = 4x.
So, 4x + 5x + 4x = 260.
13x = 260.
x = 20.
9th number = 5x = 5 × 20 = 100.

12. Option D

Profit ratio is based on capital × time.
Priyansh invested Rs. 6000 for the first 5 months and Rs. 4000 for the next 7 months.
Priyansh's capital-time = 6000 × 5 + 4000 × 7 = 30000 + 28000 = 58000.
Tarun invested Rs. 4500 for the first 5 months and Rs. 6000 for the next 7 months.
Tarun's capital-time = 4500 × 5 + 6000 × 7 = 22500 + 42000 = 64500.
Ratio of profit shares = 58000:64500 = 116:129.
Total parts = 116 + 129 = 245.
Tarun's share = 129/245 × 24500 = Rs. 12900.

CHECKLIST BY AASHISH ARORA

13. Option E

Let the initial quantity of mixture P = M ml.
Mango juice in mixture P = 30% of M = 3M/10.
Water in mixture P = 70% of M = 7M/10.
After adding 3x ml mango juice and x ml water:
Mango juice = 3M/10 + 3x.
Water = 7M/10 + x.
Final ratio = 3:4.
So, (3M/10 + 3x)/(7M/10 + x) = 3/4.
By cross multiplication:
4 × (3M/10 + 3x) = 3 × (7M/10 + x).
12M/10 + 12x = 21M/10 + 3x.
12x − 3x = 21M/10 − 12M/10.
9x = 9M/10.
x = M/10.
So, M/x = 10.

14. Option C

Ishaan is 20% more than Mohit.
So, Ishaan:Mohit = 120:100 = 6:5.

Let Ishaan's present age = 6x and Mohit's present age = 5x.
Ishaan is 50% less than Karan, so Ishaan is 50% of Karan.
Therefore, Karan's present age = 12x.
Given ratio:
(Mohit's age after 17 years):(Karan's age 6 years ago) = 2:3.
So, $(5x + 17):(12x - 6) = 2:3$.
$3 \times (5x + 17) = 2 \times (12x - 6)$.
$15x + 51 = 24x - 12$.
$63 = 9x$.
$x = 7$.
Ishaan's age = $6 \times 7 = 42$ years.
Karan's age = $12 \times 7 = 84$ years.
Required difference = $84 - 42 = 42$ years.

15. Option A
Let the speed of the boat in still water = 5x km/hr.
Downstream speed is 20% more than still water speed.
So, downstream speed = 6x km/hr.
Stream speed = $6x - 5x = x$ km/hr.
Upstream speed = $5x - x = 4x$ km/hr.
Given total time:
$144/(4x) + 216/(6x) = 12$.
$36/x + 36/x = 12$.
$72/x = 12$.
$x = 6$.
Upstream speed = $4x = 24$ km/hr.
Downstream speed = $6x = 36$ km/hr.
Time to cover 72 km upstream = $72 \div 24 = 3$ hours.
Time to cover 72 km downstream = $72 \div 36 = 2$ hours.
Total time = $3 + 2 = 5$ hours.

16. Option C

Let the present age of Saurabh = y years.
Present age of Nikhil = 38 years
p years from now:
Age of Nikhil = $38 + p$
Age of Saurabh = $y + p$

Given ratio = 10:7
So, $(38 + p)/(y + p) = 10/7$
$7(38 + p) = 10(y + p)$
$266 + 7p = 10y + 10p$
$10y = 266 - 3p$ ...(1)
After $(p + 5)$ years from now:
Age of Nikhil = $38 + p + 5 = 43 + p$
Age of Saurabh = $y + p + 5$
Given ratio = 15:11
So, $(43 + p)/(y + p + 5) = 15/11$
$11(43 + p) = 15(y + p + 5)$
$473 + 11p = 15y + 15p + 75$
$15y = 398 - 4p$ ...(2)
From equation (1), $10y = 266 - 3p$
Multiplying by 3/2:
$15y = 399 - 4.5p$
Now compare with equation (2):
$399 - 4.5p = 398 - 4p$
$1 = 0.5p$
$p = 2$
Put $p = 2$ in equation (1):
$10y = 266 - 3 \times 2$
$10y = 260$
$y = 26$

17. Option D

Let the HCF of the two numbers = h.
Let the two numbers be ha and hb, where a and b are co-prime.
LCM = hab
Given, LCM = 24 × HCF
$hab = 24h$
$ab = 24$
Product of two numbers = HCF × LCM
$21600 = h \times 24h$
$21600 = 24h^2$
$h^2 = 21600/24 = 900$
$h = 30$
Now, $ab = 24$.

Co-prime factor pairs of 24 are 1 and 24, and 3 and 8.
For maximum difference, take the pair 1 and 24.
Numbers = 30 × 1 and 30 × 24
Numbers = 30 and 720
Maximum difference = 720 − 30 = 690

18. Option B

Speed of passenger train = 72 km/hr
Convert into m/s:
72 × 5/18 = 20 m/s
Time taken to cross a pole = 12 seconds
Length of passenger train = 20 × 12 = 240 metres
Speed of goods train = 54 km/hr
Relative speed in opposite direction = 72 + 54 = 126 km/hr
Convert into m/s:
126 × 5/18 = 35 m/s
Time taken to cross each other = 18 seconds
Total length of both trains = 35 × 18 = 630 metres
Length of goods train = 630 − 240 = 390 metres

19. Option E

Let the cost price of the article = Rs. x.
Profit when sold for Rs. 1700 = 1700 − x
Loss when sold for Rs. 800 = x − 800
Given, profit = 2 × loss
1700 − x = 2(x − 800)
1700 − x = 2x − 1600
3300 = 3x
x = 1100
Cost price = Rs. 1100
Required selling price for 40% profit = 140% of cost price
Required selling price = 140/100 × 1100
Required selling price = Rs. 1540

20. Option B

Let the efficiency of Tushar = 5 units/day.
Raghav is 20% more efficient than Tushar.

Efficiency of Raghav = 5 × 120/100 = 6 units/day
Prakash is 50% less efficient than Raghav.
Efficiency of Prakash = 6 × 50/100 = 3 units/day
So, efficiency ratio of Prakash, Raghav and Tushar = 3:6:5
Together efficiency = 3 + 6 + 5 = 14 units/day
They complete the work in 42 days.
Total work = 14 × 42 = 588 units
Work done by Raghav and Tushar in 36 days:
Efficiency of Raghav and Tushar = 6 + 5 = 11 units/day
Work done = 11 × 36 = 396 units
Remaining work = 588 − 396 = 192 units
Time taken by Prakash alone = 192/3 = 64 days

CHECKLIST BY AASHISH ARORA

3. Quadratic Equations

CHECKLIST BY AASHISH ARORA

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

A. if $x > y$

B. if $x \geq y$

C. if $x < y$

D. if $x \leq y$

E. if $x = y$ or relationship cannot be established

1.) I. $3x^2 - 2x - 8 = 0$

II. $5y^2 + 18y + 16 = 0$

2.) I. $2x^2 - 12\sqrt{7}x + 70 = 0$

II. $10y^2 - 33y - 72 = 0$

3.) I. $7x^2 + 26x - 8 = 0$

II. $14y^2 - 72y + 88 = 0$

4.) I. $x^2 - 8\sqrt{5}x + 60 = 0$

II. $2y^2 - 6\sqrt{5}y + 20 = 0$

5.) I. $5x^2 + 44x + 87 = 0$

II. $y^2 - 8y - 33 = 0$

6.) I. $2x^2 - 11x - 63 = 0$

II. $8y^2 + 8y - 30 = 0$

7.) I. $3x^2 - 21\sqrt{2}x + 60 = 0$

II. $10y^2 + 19y - 12 = 0$

8.) I. $14x^2 - 16x - 30 = 0$

II. $4y^2 + 34y + 30 = 0$

9.) I. $7x^2 - 58x + 115 = 0$

II. $2y^2 - 12\sqrt{7}y + 126 = 0$

10.) I. $3x^2 - 35x + 98 = 0$

II. $6y^2 - 7y - 49 = 0$

11.) I. $5x^2 - 20\sqrt{5}x + 100 = 0$

II. $2y^2 - 16\sqrt{5}y + 120 = 0$

12.) I. $6x^2 - 23x - 35 = 0$

II. $10y^2 - 68y + 114 = 0$

13.) I. $9x^2 + 69x + 130 = 0$

II. $6y^2 - 32y - 88 = 0$

14.) I. $x^2 - 8\sqrt{7}x + 105 = 0$

II. $7y^2 - 67y + 88 = 0$

15.) I. $5x^2 - 38x + 33 = 0$

II. $10y^2 + 22y - 32 = 0$

16.) I. $x^2 - 8\sqrt{6}x + 72 = 0$

II. $12y^2 - 58y + 60 = 0$

17.) I. $12x^2 - 58x + 56 = 0$

II. $10y^2 - 73y + 133 = 0$

18.) I. $5x^2 - 44x + 95 = 0$

II. $4y^2 - 52y + 133 = 0$

19.) I. $2x^2 - 4\sqrt{5}x + 10 = 0$

II. $4y^2 - 38y + 88 = 0$

20.) I. $2x^2 - 30x + 112 = 0$

II. $5y^2 - 14y - 87 = 0$

Answers:

1. A
2. E
3. C
4. B
5. D
6. E
7. A
8. B
9. C
10. A
11. D
12. E
13. C
14. E
15. B
16. A
17. D
18. E
19. C
20. A

Solutions:

(1) $x = -\frac{4}{3}, 2$

$y = -2, -\frac{8}{5}$

(2) $x = \sqrt{7}, 5\sqrt{7}$

$y = -\frac{3}{2}, \frac{24}{5}$

(3) $x = -4, \frac{2}{7}$

$y = 2, \frac{22}{7}$

(4) $x = 2\sqrt{5}, 6\sqrt{5}$

$y = \sqrt{5}, 2\sqrt{5}$

(5) $x = -\frac{29}{5}, -3$

$y = -3, 11$

(6) $x = -\frac{7}{2}, 9$

$y = -\frac{5}{2}, \frac{3}{2}$

(7) $x = 2\sqrt{2}, 5\sqrt{2}$

$y = -\frac{12}{5}, \frac{1}{2}$

(8) $x = -1, \frac{15}{7}$

$y = -\frac{15}{2}, -1$

(9) $x = \frac{23}{7}, 5$

$y = 3\sqrt{7}, 3\sqrt{7}$

(10) $x = \frac{14}{3}, 7$

$y = -\frac{7}{3}, \frac{7}{2}$

(11) $x = 2\sqrt{5}, 2\sqrt{5}$

$y = 2\sqrt{5}, 6\sqrt{5}$

(12) $x = -\frac{7}{6}, 5$

$y = 3, \frac{19}{5}$

(13) $x = -\frac{13}{3}, -\frac{10}{3}$

$y = -2, \frac{22}{3}$

(14) $x = 3\sqrt{7}, 5\sqrt{7}$

$y = \frac{11}{7}, 8$

(15) $x = 1, \frac{33}{5}$

$y = -\frac{16}{5}, 1$

(16) $x = 2\sqrt{6}, 6\sqrt{6}$

$y = \frac{3}{2}, \frac{10}{3}$

(17) $x = \frac{4}{3}, \frac{7}{2}$

$y = \frac{7}{2}, \frac{19}{5}$

(18) $x = \frac{19}{5}, 5$

$y = \frac{7}{2}, \frac{19}{2}$

(19) $x = \sqrt{5}, \sqrt{5}$

$y = 4, \frac{11}{2}$

(20) $x = 7, 8$

$y = -3, \frac{29}{5}$

4. WRONG NUMBER SERIES

(1) 1840, 1871, 1908, 1971, 2044, 2135

(a) 1871

(b) 2044

(c) 1908

(d) 2135

(e) None of these

(2) 118, 259, 778, 3135, 15704, 94255

(a) 259

(b) 778

(c) 15704

(d) 94255

(e) None of these

(3) 48952, 24468, 8144, 2016, 384, 40

(a) 48952

(b) 24468

(c) 384

(d) 40

(e) None of these

(4) 2200, 2184, 2139, 2081, 2002, 1900

(a) 2139

(b) 2184

(c) 2002

(d) 1900

(e) None of these

(5) 840, 828, 812, 768, 720, 660

(a) 840

(b) 828

(c) 812

(d) 720

(e) None of these

(6) 1250, 1428, 1644, 1933, 2294, 2735

(a) 1428

(b) 1644

(c) 1933

(d) 2735

(e) None of these

(7) 720, 726, 750, 798, 960, 1446

(a) 726

(b) 960

(c) 750

(d) 1446

(e) None of these

(8) 256, 376, 552, 820, 1198, 1773

(a) 256

(b) 376

(c) 1198

(d) 1773

(e) None of these

(9) 489, 539, 591, 653, 701, 759

(a) 591

(b) 653

(c) 759

(d) 539

(e) None of these

(10) 900, 917, 940, 974, 1004, 1045

(a) 940

(b) 974

(c) 1045

(d) 917

(e) None of these

(11) 2500, 2460, 2402, 2330, 2240, 2130

(a) 2460

(b) 2330

(c) 2240

(d) 2130

(e) None of these

(12) 1000, 1012, 1132, 1128, 1278, 1530

(a) 1000

(b) 1012

(c) 1132

(d) 1278

(e) None of these

(13) 620.25, 634, 647.5, 666.75, 689.75, 716.5

(a) 620.25

(b) 634

(c) 689.75

(d) 716.5

(e) None of these

(14) 4000, 3955, 3865, 3695, 3325, 2605

(a) 3955

CHECKLIST BY AASHISH ARORA

(b) 3865

(c) 3325

(d) 2605

(e) None of these

(15) 145, 422, 1249, 3728, 11170, 33454

(a) 145

(b) 1249

(c) 33454

(d) 11170

(e) None of these

(16) 774, 396, 220, 132, 102, 100

(a) 396

(b) 102

(c) 220

(d) 774

(e) None of these

(17) 880, 910, 968, 1090, 1302, 1640

(a) 910

(b) 1090

(c) 968

(d) 1640

(e) None of these

(18) 180, 369, 125, 505, 107, 649

(a) 505

(b) 369

(c) 107

(d) 649

(e) None of these

(19) 210, 413, 812, 1595, 3178, 6321

(a) 812

(b) 1595

(c) 3178

(d) 6321

(e) None of these

(20) 120, 245, 480, 982, 1930, 3885

(a) 120

(b) 245

(c) 1930

(d) 3885

(e) None of these

Answers

(1) c

(2) a

(3) e

(4) b

(5) c

CHECKLIST BY AASHISH ARORA

(6) a

(7) c

(8) e

(9) b

(10) b

(11) a

(12) c

(13) b

(14) e

(15) d

(16) c

(17) a

(18) b

(19) b

(20) e

Solutions

(1) +31, +43, +57, +73, +91; second differences: +12, +14, +16, +18.

(2) ×2 + 17, ×3 + 19, ×4 + 23, ×5 + 29, ×6 + 31.

(3) ÷2 − 8, ÷3 − 12, ÷4 − 16, ÷5 − 20, ÷6 − 24.

(4) −22, −39, −58, −79, −102; second differences: −17, −19, −21, −23.

(5) −12, −24, −36, −48, −60.

(6) $+13^2, +15^2, +17^2, +19^2, +21^2$.

(7) +6, +18, +54, +162, +486.

(8) ×1.5 − 8, ×1.5 − 12, ×1.5 − 16, ×1.5 − 20, ×1.5 − 24.

(9) +50, +52, +54, +56, +58.

(10) +17, +23, +29, +35, +41; second differences: +6, +6, +6, +6.

(11) −(6 × 7), −(7 × 8), −(8 × 9), −(9 × 10), −(10 × 11).

(12) $+(2^3 + 2^2), +(3^3 + 3^2), +(4^3 + 4^2), +(5^3 + 5^2), +(6^3 + 6^2)$.

(13) +11.75, +15.5, +19.25, +23, +26.75; second differences: +3.75, +3.75, +3.75, +3.75.

(14) −45, −90, −180, −360, −720.

(15) ×3 − 13, ×3 − 17, ×3 − 19, ×3 − 23, ×3 − 29.

(16) $\div 2 + 3^2, \div 2 + 4^2, \div 2 + 5^2, \div 2 + 6^2, \div 2 + 7^2$.

(17) $+(3^3 - 1), +(4^3 - 2), +(5^3 - 3), +(6^3 - 4), +(7^3 - 5)$.

(18) ×2 + 3, ÷3 + 4, ×4 + 5, ÷5 + 6, ×6 + 7.

(19) ×2 − 7, ×2 − 14, ×2 − 21, ×2 − 28, ×2 − 35.

(20) ×2 + 5, ×2 − 10, ×2 + 15, ×2 − 20, ×2 + 25.

CHECKLIST BY AASHISH ARORA

5. MISSING NUMBER SERIES

CHECKLIST BY AASHISH ARORA

(1) 7450, 7436, 7410, ?, 7290, 7180

(a) 7360
(b) 7364
(c) 7368
(d) 7372
(e) None of these

(2) 8, 19, 63, ?, 1317, 7917

(a) 257
(b) 259
(c) 261
(d) 263
(e) None of these

(3) 42, 58, 87, ?, 200, 292

(a) 133
(b) 135
(c) 137
(d) 139
(e) None of these

(4) 55, 86, ?, 239, 396, 651

(a) 141
(b) 145
(c) 147
(d) 149
(e) None of these

(5) 6854, 3432, ?, 290, 60, 11

(a) 1142
(b) 1144
(c) 1148
(d) 1152
(e) None of these

(6) 76.8, 80.4, 87.6, ?, 112.8, 130.8

(a) 97.4
(b) 98.4
(c) 99.4
(d) 100.4
(e) None of these

(7) 65, 81, 117, ?, 281, 425

(a) 181
(b) 183
(c) 185
(d) 187
(e) None of these

(8) 980, 974, 959, ?, 886, 820

(a) 927
(b) 929
(c) 931

(d) 933
(e) None of these

(9) 15, 37, 77, ?, 235, 365

(a) 141
(b) 143
(c) 145
(d) 147
(e) None of these

(10) 64, 256, ?, 520, 262, 1048

(a) 128
(b) 130
(c) 132
(d) 134
(e) None of these

(11) 1500, 1512, 1548, ?, 1778, 2030

(a) 1626
(b) 1628
(c) 1630
(d) 1632
(e) None of these

(12) 21, 46, 101, ?, 461, 958

(a) 216
(b) 218
(c) 220
(d) 222
(e) None of these

(13) 6400, 1600, 3200, ?, 2400, 600

(a) 800
(b) 810
(c) 820
(d) 830
(e) None of these

(14) 146, 162, 186, ?, 290, 394

(a) 220
(b) 222
(c) 224
(d) 226
(e) None of these

(15) 2400, 2436, 2508, ?, 2760, 2940

(a) 2616
(b) 2618
(c) 2620
(d) 2622
(e) None of these

(16) 4, 16, 56, ?, 556, 1688

(a) 176
(b) 180
(c) 184
(d) 188
(e) None of these

(17) 980, 961, 923, ?, 695, 391

(a) 845
(b) 847
(c) 849
(d) 851
(e) None of these

(18) 22, 37, 68, ?, 258, 513

(a) 131
(b) 133
(c) 135
(d) 137
(e) None of these

(19) 6026, 3018, ?, 260, 60, 19

(a) 1012
(b) 1014

CHECKLIST BY AASHISH ARORA

(c) 1016
(d) 1018
(e) None of these

(20) 44, 46, 52, ?, 196, 916

(a) 72
(b) 74
(c) 76
(d) 78
(e) None of these

Answers:

(1) b
(2) c
(3) a
(4) b
(5) c
(6) b
(7) a
(8) c
(9) a
(10) b
(11) b
(12) b
(13) a
(14) d
(15) a
(16) b
(17) b
(18) a
(19) a
(20) c

Solutions:

(1) $-(2^2 + 10), -(4^2 + 10), -(6^2 + 10), -(8^2 + 10), -(10^2 + 10)$

(2) $\times 2 + 3, \times 3 + 6, \times 4 + 9, \times 5 + 12, \times 6 + 15$

(3) +16, +29, +46, +67, +92

Second difference: +13, +17, +21, +25

(4) Sum of previous two numbers +4, +8, +12, +16

55 + 86 + 4 = 145

(5) $\div 2 + 5, \div 3 + 4, \div 4 + 3, \div 5 + 2, \div 6 + 1$

(6) +3.6, +7.2, +10.8, +14.4, +18

(7) $+4^2, +6^2, +8^2, +10^2, +12^2$

(8) $-2\times 3, -3\times 5, -4\times 7, -5\times 9, -6\times 11$

(9) Terms are: $n^3 + 3n + 1$, for n = 2, 3, 4, 5, 6, 7

(10) $\times 4, \div 2 + 2, \times 4, \div 2 + 2, \times 4$

(11) +12, +36, +80, +150, +252

Pattern: $+n^3 + n^2$, for n = 2, 3, 4, 5, 6

(12) $\times 2 + 2^2, \times 2 + 3^2, \times 2 + 4^2, \times 2 + 5^2, \times 2 + 6^2$

(13) $\div 4, \times 2, \div 4, \times 3, \div 4$

(14) +16, +24, +40, +64, +104

Each difference is the sum of the previous two differences.

(15) +36, +72, +108, +144, +180

(16) $\times 3 + 4, \times 3 + 8, \times 3 + 12, \times 3 + 16, \times 3 + 20$

(17) −19, −38, −76, −152, −304

(18) +15, +31, +63, +127, +255

Each difference is ×2 + 1 of the previous difference.

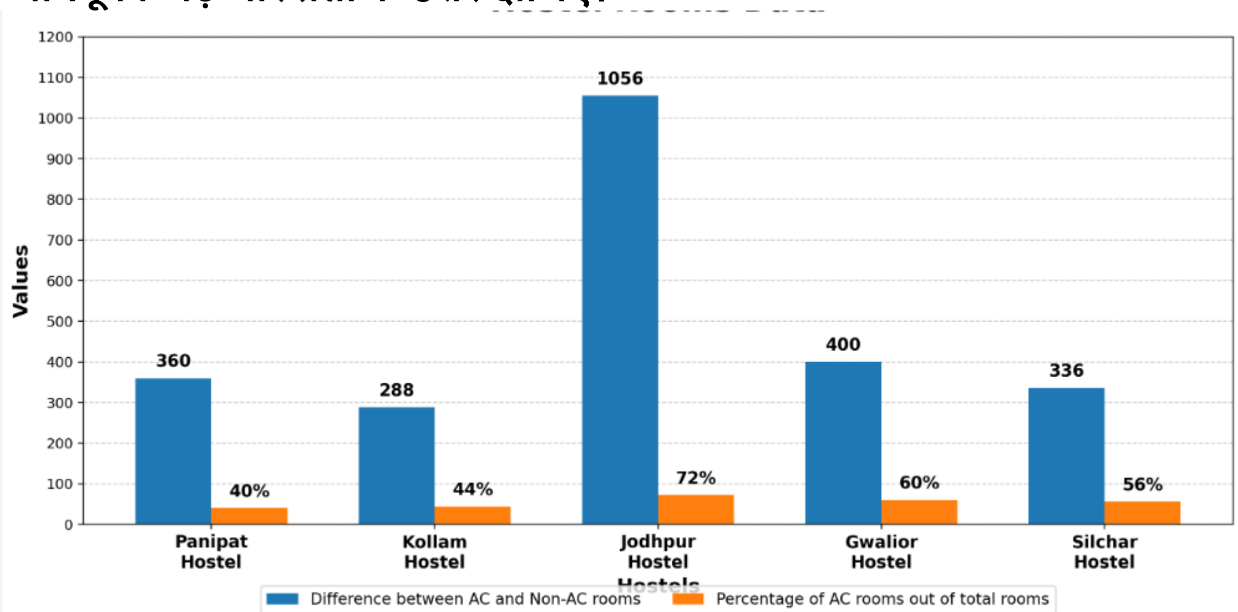
(19) ÷2 + 5, ÷3 + 6, ÷4 + 7, ÷5 + 8, ÷6 + 9

(20) +2!, +3!, +4!, +5!, +6!

CHECKLIST BY AASHISH ARORA

6. DATA INTERPRETATION

Set 1: The bar graph shows the percentage of AC rooms out of total rooms and the difference between the number of AC and Non-AC rooms in five different hostels. Total rooms = AC rooms + Non-AC rooms. All values in the data table are multiples of 4. Read the bar graph carefully and answer the questions.
दंड आलेख पाँच अलग-अलग छात्रावासों में कुल कमरों में से AC कमरों का प्रतिशत और AC तथा Non-AC कमरों की संख्या के बीच का अंतर दर्शाता है। कुल कमरे = AC कमरे + Non-AC कमरे। डेटा तालिका के सभी मान 4 के गुणज हैं। दंड आलेख को ध्यानपूर्वक पढ़ें और प्रश्नों के उत्तर दीजिए।



Q1: Find the average number of AC rooms in Panipat Hostel and Kollam Hostel together.
पानीपत छात्रावास और कोल्लम छात्रावास में AC कमरों की औसत संख्या ज्ञात कीजिए।
(A) 864
(B) 876
(C) 888
(D) 912
(E) 924

Q2: Each AC room and each Non-AC room in Jodhpur Hostel can accommodate 8 students and 4 students, respectively. Out of these, at least 25% students in each AC room and 75% students in each Non-AC room must be females. Find the minimum number of female students who must be accommodated in all rooms of Jodhpur Hostel.

जोधपुर छात्रावास में प्रत्येक AC कमरा और प्रत्येक Non-AC कमरा क्रमशः 8 और 4 छात्रों को समायोजित कर सकता है। इनमें से प्रत्येक AC कमरे में कम से कम 25% छात्राएँ और प्रत्येक Non-AC कमरे में 75% छात्राएँ होनी चाहिए। जोधपुर छात्रावास के सभी कमरों में समायोजित की जाने वाली न्यूनतम छात्राओं की संख्या ज्ञात कीजिए।

(A) 5472
(B) 5376
(C) 5568
(D) 5280
(E) 5664

CHECKLIST BY AASHISH ARORA

Q3: Total rooms in Kollam Hostel and Silchar Hostel together are what percentage of the number of Non-AC rooms in Gwalior Hostel?

कोल्लम छात्रावास और सिलचर छात्रावास में मिलाकर कुल कमरे, ग्वालियर छात्रावास में Non-AC कमरों की संख्या का कितना प्रतिशत हैं?

(A) 600%
(B) 625%
(C) 675%
(D) 700%
(E) 650%

Q4: In Kollam Hostel, the monthly maintenance charge for each Non-AC room is Rs. 5, while in Silchar Hostel, the monthly maintenance charge for each Non-AC room is Rs. 12. Find the difference between the total monthly maintenance revenue from Non-AC rooms in Kollam Hostel and Silchar Hostel.

कोल्लम छात्रावास में प्रत्येक Non-AC कमरे का मासिक रखरखाव शुल्क 5 रुपये है, जबकि सिलचर छात्रावास में प्रत्येक Non-AC कमरे का मासिक रखरखाव शुल्क 12 रुपये है। कोल्लम छात्रावास और सिलचर छात्रावास में Non-AC कमरों से प्राप्त कुल मासिक रखरखाव आय के बीच अंतर ज्ञात कीजिए।

(A) Rs. 7464
(B) Rs. 8064
(C) Rs. 7864
(D) Rs. 8264

(E) Rs. 8464

Q5: If the number of Non-AC rooms in Jodhpur Hostel is 2p and the number of AC rooms in Kollam Hostel is 3q, then find the value of p + q.
यदि जोधपुर छात्रावास में Non-AC कमरों की संख्या 2p है और कोल्लम छात्रावास में AC कमरों की संख्या 3q है, तो p + q का मान ज्ञात कीजिए।
(A) 640
(B) 672
(C) 680
(D) 688
(E) 696

Solutions:

Hostel	Difference	AC %	Total rooms	AC rooms	Non-AC rooms
Panipat Hostel	360	40%	1800	720	1080
Kollam Hostel	288	44%	2400	1056	1344
Jodhpur Hostel	1056	72%	2400	1728	672
Gwalior Hostel	400	60%	2000	1200	800
Silchar Hostel	336	56%	2800	1568	1232

1. Option C
Panipat Hostel: AC rooms = 40% of total
Difference = 60% − 40% = 20% of total = 360
Total rooms = 360 / 20% = 1800
AC rooms in Panipat Hostel = 1800 × 40% = 720
Kollam Hostel: Difference = 56% − 44% = 12% of total = 288
Total rooms = 288 / 12% = 2400
AC rooms in Kollam Hostel = 2400 × 44% = 1056
Average = (720 + 1056) / 2 = 888

2. Option A
Jodhpur Hostel: Difference = 72% − 28% = 44% of total = 1056
Total rooms = 1056 / 44% = 2400
AC rooms = 2400 × 72% = 1728
Non-AC rooms = 2400 − 1728 = 672
Female students in each AC room = 25% of 8 = 2
Female students in each Non-AC room = 75% of 4 = 3
Minimum female students = 1728 × 2 + 672 × 3

Minimum female students = 3456 + 2016 = 5472

3. Option E
Total rooms in Kollam Hostel = 288 / 12% = 2400
Total rooms in Silchar Hostel = 336 / 12% = 2800
Total rooms together = 2400 + 2800 = 5200
Gwalior Hostel: Difference = 60% − 40% = 20% of total = 400
Total rooms in Gwalior Hostel = 400 / 20% = 2000
Non-AC rooms in Gwalior Hostel = 2000 × 40% = 800
Required percentage = 5200 / 800 × 100 = 650%

4. Option B
Non-AC rooms in Kollam Hostel = 2400 × 56% = 1344
Non-AC rooms in Silchar Hostel = 2800 × 44% = 1232
Revenue from Kollam Hostel = 1344 × 5 = Rs. 6720
Revenue from Silchar Hostel = 1232 × 12 = Rs. 14784
Difference = 14784 − 6720 = Rs. 8064

5. Option D
Non-AC rooms in Jodhpur Hostel = 2400 × 28% = 672
2p = 672
p = 336
AC rooms in Kollam Hostel = 1056
3q = 1056
q = 352
p + q = 336 + 352 = 688

CHECKLIST BY AASHISH ARORA

Set 2: The table shows data about five mentors who prepared mock-test questions for an online learning platform. It gives the number of sessions conducted, total minutes spent, and the average number of questions prepared per session. Some data is missing. Read the data carefully and answer the following questions.

तालिका पाँच मेंटर्स द्वारा एक ऑनलाइन लर्निंग प्लेटफॉर्म के लिए तैयार किए गए मॉक-टेस्ट प्रश्नों का डेटा दर्शाती है। इसमें सेशन की संख्या, कुल मिनट और प्रति सेशन तैयार किए गए औसत प्रश्न दिए गए हैं। कुछ डेटा लुप्त है। डेटा को ध्यानपूर्वक पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।

Note: Productivity rate = (Total questions prepared / Total minutes spent) x 100.
Total questions prepared = Average questions prepared per session x Number of sessions conducted.

नोट: उत्पादकता दर = (कुल तैयार प्रश्न / कुल मिनट) x 100.
कुल तैयार प्रश्न = प्रति सेशन औसत प्रश्न x सेशन की संख्या।

The total minutes spent by Saurabh is the average of the total minutes spent by Ritika and Vinit. The productivity rate of Manasi is 20% more than that of Pooja. The average questions prepared by Ritika per session is twice that of Vinit, and Ritika's productivity rate is 150%. The productivity rate of Pooja is 125%. The number of sessions conducted by Manasi is 13 more than that of Saurabh.

सौरभ द्वारा लगाए गए कुल मिनट, रितिका और विनीत द्वारा लगाए गए कुल मिनटों का औसत हैं। मानसी की उत्पादकता दर, पूजा की उत्पादकता दर से 20% अधिक है। रितिका द्वारा प्रति सेशन तैयार किए गए औसत प्रश्न, विनीत के औसत प्रश्नों के दोगुने हैं, और रितिका की उत्पादकता दर 150% है। पूजा की उत्पादकता दर 125% है। मानसी द्वारा लिए गए सेशन की संख्या, सौरभ से 13 अधिक है।

Mentor (मेंटोर)	Number of sessions (सेशन की संख्या)	Total minutes spent (कुल मिनट)	Average questions per session (प्रति सेशन औसत प्रश्न)
Ritika (रितिका)	--	1040	--
Saurabh (सौरभ)	39	--	65
Manasi (मानसी)	--	1560	--
Vinit (विनीत)	26	520	39
Pooja (पूजा)	52	--	65

Q1: Find the total minutes spent by Pooja.
पूजा द्वारा लगाए गए कुल मिनट ज्ञात कीजिए।
(A) 2600

(B) 2704
(C) 2808
(D) 2912
(E) 2652

Q2: Find the average number of questions prepared per session by Manasi.
मानसी द्वारा प्रति सेशन तैयार किए गए औसत प्रश्नों की संख्या ज्ञात कीजिए।
(A) 39
(B) 52
(C) 65
(D) 45
(E) 56

Q3: Find the number of sessions conducted by Ritika.
रितिका द्वारा लिए गए सेशन की संख्या ज्ञात कीजिए।
(A) 18
(B) 24
(C) 20
(D) 26
(E) 22

Q4: What is the productivity rate of Vinit?
विनीत की उत्पादकता दर कितनी है?
(A) 180%
(B) 210%
(C) 225%
(D) 200%
(E) 195%

Q5: What is the productivity rate of Saurabh?
सौरभ की उत्पादकता दर कितनी है?
(A) 325%
(B) 300%
(C) 280%
(D) 250%
(E) 350%

CHECKLIST BY AASHISH ARORA

Solutions:

Mentor (मेंटोर)	Sessions (सेशन)	Total Minutes (कुल मिनट)	Average Questions (औसत प्रश्न)	Total Questions (कुल प्रश्न)	Productivity Rate (उत्पादकता दर)
Ritika (रितिका)	20	1040	78	1560	150%
Saurabh (सौरभ)	39	780	65	2535	325%
Manasi (मानसी)	52	1560	45	2340	150%
Vinit (विनीत)	26	520	39	1014	195%
Pooja (पूजा)	52	2704	65	3380	125%

1. Option B
Total questions prepared by Pooja = 52 x 65 = 3380
Pooja's productivity rate = 125%
125% = 5/4
Total minutes spent by Pooja = 3380 x 4/5
Total minutes spent by Pooja = 2704

2. Option D
Pooja's productivity rate = 125%
Manasi's productivity rate = 20% more than 125% = 150%
Total questions prepared by Manasi = 150% of 1560
Total questions prepared by Manasi = 2340
Number of sessions conducted by Manasi = 39 + 13 = 52
Average questions prepared by Manasi = 2340 / 52
Average questions prepared by Manasi = 45

3. Option C
Average questions prepared by Vinit = 39
Average questions prepared by Ritika = 2 x 39 = 78
Ritika's productivity rate = 150%
Total questions prepared by Ritika = 150% of 1040
Total questions prepared by Ritika = 1560
Number of sessions conducted by Ritika = 1560 / 78
Number of sessions conducted by Ritika = 20

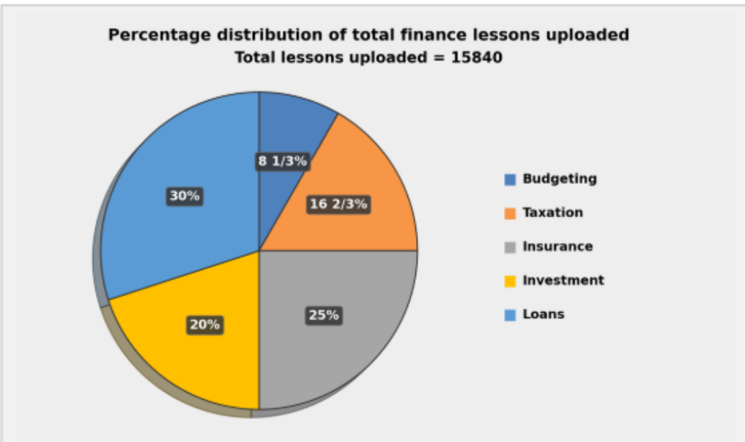
4. Option E
Total questions prepared by Vinit = 26 x 39 = 1014
Total minutes spent by Vinit = 520
Productivity rate of Vinit = 1014 / 520 x 100
Productivity rate of Vinit = 195%

5. Option A
Total minutes spent by Saurabh = (1040 + 520) / 2
Total minutes spent by Saurabh = 780
Total questions prepared by Saurabh = 39 x 65 = 2535
Productivity rate of Saurabh = 2535 / 780 x 100
Productivity rate of Saurabh = 325%

CHECKLIST BY AASHISH ARORA

Set 3: The pie chart shows the percentage distribution of the total number of finance lessons uploaded on ArthSetu Learning App in five different modules. The table shows the ratio of Hindi-language lessons to English-language lessons uploaded in each module. The total number of lessons uploaded in all five modules together is 15840. Read the data carefully and answer the following questions.

पाई चार्ट अर्थसेतु लर्निंग ऐप पर पाँच अलग-अलग मॉड्यूल में अपलोड किए गए कुल वित्तीय लेसन का प्रतिशत वितरण दर्शाता है। तालिका प्रत्येक मॉड्यूल में अपलोड किए गए हिंदी भाषा लेसन और अंग्रेज़ी भाषा लेसन का अनुपात दर्शाती है। पाँचों मॉड्यूल में मिलाकर कुल 15840 लेसन अपलोड किए गए हैं। आँकड़ों को ध्यान से पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।



The table shows the ratio of lessons uploaded in Hindi to English.

Module	Hindi : English
Budgeting	05 : 06
Taxation	08 : 07
Insurance	04 : 05
Investment	11 : 13
Loans	19 : 17

Q1: Find the ratio of the number of Hindi-language lessons uploaded in Insurance to the number of English-language lessons uploaded in Budgeting.
बीमा में अपलोड किए गए हिंदी भाषा लेसन की संख्या का बजटिंग में अपलोड किए गए अंग्रेज़ी भाषा लेसन की संख्या से अनुपात ज्ञात कीजिए।
(A) 11 : 6
(B) 17 : 9
(C) 19 : 8
(D) 22 : 9

(E) 25 : 11

Q2: The number of English-language lessons uploaded in Taxation is how much percent less than the number of Hindi-language lessons uploaded in Insurance?
कराधान में अपलोड किए गए अंग्रेज़ी भाषा लेसन की संख्या, बीमा में अपलोड किए गए हिंदी भाषा लेसन की संख्या से कितने प्रतिशत कम है?
(A) 30%
(B) 25%
(C) 37.5%
(D) 20%
(E) 33 1/3%

Q3: Find the average number of English-language lessons uploaded in Investment and Loans.
निवेश और ऋण में अपलोड किए गए अंग्रेज़ी भाषा लेसन की औसत संख्या ज्ञात कीजिए।
(A) 1896
(B) 2040
(C) 1980
(D) 1924
(E) 2100

Q4: 3/5 of the Hindi-language lessons uploaded in Budgeting and 4/11 of the Hindi-language lessons uploaded in Investment were marked as advanced lessons. Find the total number of advanced lessons from these two modules.
बजटिंग में अपलोड किए गए हिंदी भाषा लेसन के 3/5 भाग और निवेश में अपलोड किए गए हिंदी भाषा लेसन के 4/11 भाग को एडवांस लेसन के रूप में चिह्नित किया गया। इन दोनों मॉड्यूल से कुल एडवांस लेसन की संख्या ज्ञात कीजिए।
(A) 840
(B) 912
(C) 864
(D) 936
(E) 888

Q5: Out of the total Taxation lessons uploaded, 40% were uploaded by male trainers and the rest by female trainers. If 5/8 of the Hindi-language lessons uploaded in Taxation were uploaded by female trainers, then find the number of English-language Taxation lessons uploaded by male trainers.

CHECKLIST BY AASHISH ARORA

कराधान में अपलोड किए गए कुल लेसन में से 40% पुरुष प्रशिक्षकों द्वारा और शेष महिला प्रशिक्षकों द्वारा अपलोड किए गए। यदि कराधान में अपलोड किए गए हिंदी भाषा लेसन के 5/8 भाग महिला प्रशिक्षकों द्वारा अपलोड किए गए, तो पुरुष प्रशिक्षकों द्वारा अपलोड किए गए अंग्रेज़ी भाषा कराधान लेसन की संख्या ज्ञात कीजिए।

(A) 496

(B) 528

(C) 560

(D) 512

(E) None of these

CHECKLIST

Solutions:

Module	Hindi	English	Total
Budgeting	600	720	1320
Taxation	1408	1232	2640
Insurance	1760	2200	3960
Investment	1452	1716	3168
Loans	2508	2244	4752

Common Calculation:

Total lessons uploaded = 15840

Budgeting = 8 1/3% of 15840 = 1320

Taxation = 16 2/3% of 15840 = 2640

Insurance = 25% of 15840 = 3960

Investment = 20% of 15840 = 3168

Loans = 30% of 15840 = 4752

AASHISH ARORA

1. Option D

Insurance Hindi-language lessons = 1760

Budgeting English-language lessons = 720

Required ratio = 1760 : 720 = 22 : 9

2. Option A

Taxation English-language lessons = 1232

Insurance Hindi-language lessons = 1760

Difference = 1760 − 1232 = 528

Required percentage = 528/1760 × 100 = 30%

3. Option C
Investment English-language lessons = 1716
Loans English-language lessons = 2244
Required average = (1716 + 2244)/2 = 1980

4. Option E
Advanced Budgeting lessons = 3/5 of 600 = 360
Advanced Investment lessons = 4/11 of 1452 = 528
Total advanced lessons = 360 + 528 = 888

5. Option B
Total Taxation lessons = 2640
Male trainers uploaded = 40% of 2640 = 1056
Female trainers uploaded = 2640 − 1056 = 1584
Female trainers uploaded Hindi lessons = 5/8 of 1408 = 880
Female trainers uploaded English lessons = 1584 − 880 = 704
Total English Taxation lessons = 1232
Male trainers uploaded English lessons = 1232 − 704 = 528

CHECKLIST BY AASHISH ARORA

Set 4: Three sports academies, Sonal, Rishi and Lavanya, sold two types of training passes: Swimming Passes and Badminton Passes. The ratio of Swimming Passes sold by Sonal to Swimming Passes sold by Lavanya is 5 : 8. The number of Badminton Passes sold by Rishi is 126 more than the number of Badminton Passes sold by Sonal. The number of Swimming Passes sold by Rishi is (m + 49) more than the number of Swimming Passes sold by Sonal. The total number of passes sold by Rishi and Lavanya is (10m – 70) and (9m – 35), respectively. The number of Badminton Passes sold by Lavanya is (5m – 35). The average number of Badminton Passes sold by all three academies is 371.

तीन स्पोर्ट्स अकादमियों, सोनल, ऋषि और लावण्या ने दो प्रकार के ट्रेनिंग पास बेचे: स्विमिंग पास और बैडमिंटन पास। सोनल द्वारा बेचे गए स्विमिंग पास और लावण्या द्वारा बेचे गए स्विमिंग पास का अनुपात 5 : 8 है। ऋषि द्वारा बेचे गए बैडमिंटन पास की संख्या, सोनल द्वारा बेचे गए बैडमिंटन पास से 126 अधिक है। ऋषि द्वारा बेचे गए स्विमिंग पास की संख्या, सोनल द्वारा बेचे गए स्विमिंग पास से (m + 49) अधिक है। ऋषि और लावण्या द्वारा बेचे गए कुल पास क्रमशः (10m – 70) और (9m – 35) हैं। लावण्या द्वारा बेचे गए बैडमिंटन पास की संख्या (5m – 35) है। तीनों अकादमियों द्वारा बेचे गए बैडमिंटन पास की औसत संख्या 371 है।

Q1: The total number of Badminton Passes sold by Sonal and Lavanya together is what percentage more or less than the number of Swimming Passes sold by Rishi?
सोनल और लावण्या द्वारा मिलाकर बेचे गए बैडमिंटन पास की कुल संख्या, ऋषि द्वारा बेचे गए स्विमिंग पास की संख्या से कितने प्रतिशत अधिक या कम है?
(A)80% more
(B)120% more
(C)100% more
(D)90% more
(E)110% more

Q2: If the selling price of each Swimming Pass and each Badminton Pass sold by Rishi is Rs. 42 and Rs. 28, respectively, find the difference between the revenue generated from Swimming Passes and Badminton Passes sold by Rishi.
यदि ऋषि द्वारा बेचे गए प्रत्येक स्विमिंग पास और प्रत्येक बैडमिंटन पास का विक्रय मूल्य क्रमशः 42 रुपये और 28 रुपये है, तो ऋषि द्वारा बेचे गए स्विमिंग पास और बैडमिंटन पास से प्राप्त राजस्व के बीच अंतर ज्ञात कीजिए।
(A)Rs. 2240
(B)Rs. 2310
(C)Rs. 2380

(D)Rs. 2520
(E)Rs. 2450

Q3: If the number of Badminton Passes sold by Manav is 77 less than that sold by Sonal, and the total number of passes sold by Manav is (8m + 35), then find the number of Swimming Passes sold by Manav.
यदि मानव द्वारा बेचे गए बैडमिंटन पास की संख्या, सोनल द्वारा बेचे गए बैडमिंटन पास से 77 कम है, और मानव द्वारा बेचे गए कुल पास (8m + 35) हैं, तो मानव द्वारा बेचे गए स्विमिंग पास की संख्या ज्ञात कीजिए।
(A)455
(B)483
(C)504
(D)476
(E)490

Q4: In Sonal Academy, the number of invalid Swimming Passes is twice the number of invalid Badminton Passes, and the ratio of valid Swimming Passes to valid Badminton Passes is 10 : 19. Find the total number of valid passes sold by Sonal.
सोनल अकादमी में अमान्य स्विमिंग पास की संख्या, अमान्य बैडमिंटन पास की संख्या की दोगुनी है, और वैध स्विमिंग पास तथा वैध बैडमिंटन पास का अनुपात 10 : 19 है। सोनल द्वारा बेचे गए कुल वैध पास की संख्या ज्ञात कीजिए।
(A)392
(B)420
(C)399
(D)406
(E)413

Q5: Find the average of the number of Badminton Passes sold by Sonal, the number of Badminton Passes sold by Lavanya, and the number of Swimming Passes sold by Rishi.
सोनल द्वारा बेचे गए बैडमिंटन पास, लावण्या द्वारा बेचे गए बैडमिंटन पास और ऋषि द्वारा बेचे गए स्विमिंग पास की औसत संख्या ज्ञात कीजिए।
(A)343
(B)336
(C)350
(D)329

CHECKLIST BY AASHISH ARORA

(E)357

Solutions:

Academy	Swimming Passes	Badminton Passes	Total Passes
Sonal	210	301	511
Rishi	343	427	770
Lavanya	336	385	721
Total	**889**	**1113**	**2002**

1. Option C

Badminton Passes sold by Sonal and Lavanya together = 301 + 385
= 686
Swimming Passes sold by Rishi = 343
Percentage more = (686 − 343) ÷ 343 × 100
= 100% more

2. Option E

Revenue from Swimming Passes sold by Rishi = 343 × 42
= Rs. 14406
Revenue from Badminton Passes sold by Rishi = 427 × 28
= Rs. 11956
Required difference = 14406 − 11956
= Rs. 2450

3. Option B

m = 84
Total passes sold by Manav = 8m + 35
= 8 × 84 + 35
= 707
Badminton Passes sold by Manav = 301 − 77
= 224
Swimming Passes sold by Manav = 707 − 224
= 483

4. Option D

Let invalid Badminton Passes = x
Invalid Swimming Passes = 2x

CHECKLIST BY AASHISH ARORA

Valid Swimming Passes = 210 − 2x
Valid Badminton Passes = 301 − x
(210 − 2x) : (301 − x) = 10 : 19
19(210 − 2x) = 10(301 − x)
3990 − 38x = 3010 − 10x
28x = 980
x = 35
Total valid passes = (210 − 70) + (301 − 35)
= 406

5. Option A

Required average = (301 + 385 + 343) ÷ 3
= 1029 ÷ 3
= 343

CHECKLIST BY AASHISH ARORA